MILITARY SPECIFICATION

KIT, CONVERSION
H4277/H4272 EXTRACTOR TOOL
M753 PROJECTILE

This amendment forms a part of Military Specification MIL-K-63185(AR) dated 25 June 1982, and is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

3.4.1 Delete entirely and substitute:

3.4.1 Life cycle tensile strength. With the forcing cone end of the extraction tool subassembly restrained, the extraction tool subassembly shall withstand a tensile force of 54,000 lb ± 10% on the opposite rod end for 25 cycles at 70° ± 20°F without breakage, cracking, distortion or any other evidence of physical damage. Testing shall be performed in accordance with the applicable conditions of 4.5.2.

PAGE 23

4.5.2 Delete entirely and substitute:

4.5.2 Tensile strength (see 3.4). Dwg. 9312672 illustrates the actual use of the extraction tool subassembly. However, for test purposes the tensile strength tests may be performed horizontally or vertically. The forcing cone end of the extraction tool subassembly shall be fully expanded in an approved and restrained test fixture simulating the internal dimensional characteristics of an M753 rocket motor boattail (see dwg. 9299050). In assembling the extractor tool subassembly into the fixture care should be exercised to assure that the bearing plate force of the subassembly presses tightly and evenly around the face of the test fixture and the jaws press evenly against the internal simulated boattail taper without tilting or cocking as the extraction tool subassembly jaws are expanded. The boattail simulation test fixture shall be designed by the contractor to withstand all the tensile forces required by this specification at the loads specified. The boattail simulation fixture or parts thereof may be expandable but the contractor shall provide sufficient test fixtures to perform all of the tensile strength tests required at the specified loads (first article and quality conformance tensile strength screening tests). Approval by the Government of the contractor designed and furnished test fixtures (AIE) shall not relieve the contractor of responsiblility for successfully complying with all the requirements of this specification and providing tooling and fixtures capable of fully and successfully performing all the tests required herein. An H4277 mating rod (dwg. 9312541) furnished

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AMSC N/A
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by the acquiring activity shall be threaded onto the forward rod end of the subassembly and secured with a retaining bolt. An approved forcing mechanism shall be secured to the mating rod. With the test fixture and the forcing cone end of the extraction tool subassembly restrained, the applicable tensile force indicated below shall be exerted on the mating rod end, retained for ten seconds minimum, and released. The tensile load shall be repeated for the number of cycles indicated. After being subjected to the required number of cycles, the H4272 extraction tool subassembly shall be visually examined for the deficiencies cited in 3.4.1 and for evidence of physical damage, including inspection of the H4272 rod thread with approved thread gages.

First Article Life Cycle Test:

54000 lb $\pm$ 10% at $70^{\circ} \pm 20^{\circ}$F (25 cycles)

Quality Conformance Tensile Strength Screening:

45000 lb $\pm$ 10% at $70^{\circ} \pm 20^{\circ}$F (3 cycles)

Custodian:
Army - AR

Preparing Activity:
Army - AR

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